

2	(a)	(i)	As r approaches zero, the potential drops lower. This corresponds to the potential near the larger mass. Therefore, r is measured from the Earth.		B1
		(ii)	1	It is the point where the gravitational field strength due to Earth is equal and opposite to the gravitational field strength due to the Moon or net gravitation field strength is zero	B1
			2	$2.25 \times 10^9 \text{ J} = m \times \Delta \phi_{\text{moon} - \text{max}}$ $= 1500 \times \Delta \phi_{\text{moon} - \text{max}}$ $\Delta \phi_{\text{moon} - \text{max}} = 1.5 \times 10^6 \text{ J kg}^{-1}$ $\phi_{\text{max}} - \phi_{\text{moon}} = \phi_{\text{max}} - (-2.5 \times 10^6) = 1.5 \times 10^6$ $\phi_{\text{max}} = -1.0 \times 10^6 \text{ J kg}^{-1}$ Allow ecf from (a)(i)	C1 A1
			3	$\frac{1}{2} m v^2 = m \times \Delta \phi_{\text{earth} - \text{max}}$	C1